

# The generalized-Lorenz gauged $A$ - $\Phi$ formulation

An implementation specification, transcribed from

Y.-L. Li, S. Sun, Q. I. Dai and W. C. Chew, *Finite Element Implementation of the Generalized-Lorenz Gauged  $A$ - $\Phi$  Formulation for Low-Frequency Circuit Modeling*, **IEEE Trans. Antennas Propag.** **64(10)**, 4355–4364, 2016,

with the continuum theory from

W. C. Chew, *Vector Potential Electromagnetics with Generalized Gauge for Inhomogeneous Media: Formulation*, **PIER** **149**, 69–84, 2014.

Both are in `APhi_Papers/` . `docs/GAUGE_CHOICE.md` is the decision record for *why* this gauge; this document is *what to build*.

**Status: specification, nothing implemented.** Equations (1)–(37) below are transcribed from the typeset paper. §7 is **my own derivation and is not in either paper** — it is flagged as such and must be checked before it is trusted.

**§11 reports a spike that measured this, and it changes §7 and §8.** The formulation conditions beautifully for dielectrics — flat across ten decades — and **badly for lossy conductors at low frequency**, where  $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$  creates a material contrast of  $1e17$ .  $\chi$  must also be position-dependent, which breaks §7's symmetrization. **Read §11 before implementing anything.**

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## 0. The convention trap — read this first

**\*\*Chew uses  $e^{-i\omega t}$ . We use  $e^{+j\omega t}$ .** His eq. (1) is  $E = i\omega A - \nabla\Phi$ ; ours is  $E = -j\omega A - \nabla\Phi$ .

every  $i$  in the paper  $\rightarrow -j$  here

Our convention is fixed by `src/element_matrix.cpp` ( $\alpha = j\omega\sigma - \omega^2\epsilon$ ) and `src/problem.cpp` ( $A(\cos \phi + j \sin \phi)$ ), and is stamped into every `.out` header.

**Nothing below is a transcription error waiting to happen except this**, so every equation in this document is already written in *our* convention, and anyone cross-checking against the paper must apply the mapping above rather than reading the signs off the page.

The purely real matrices in §5 are unaffected — they contain no `i`.

## 1. The continuum formulation, in our convention

$$\mathbf{E} = -j\omega\mathbf{A} - \nabla\Phi \quad (1)$$

$$\nabla \cdot (\epsilon \nabla \Phi) + \chi \omega^2 \Phi = -\rho \quad (2)$$

$$\nabla \times (\mathbf{v} \nabla \times \mathbf{A}) - \omega^2 \epsilon \mathbf{A} - \epsilon \nabla [ (1/\chi) \nabla \cdot (\epsilon \mathbf{A}) ] = \mathbf{J} \quad (3)$$

$$\chi = \alpha \mu \epsilon^2 \quad (4)$$

$$(1/\chi) \nabla \cdot (\epsilon \mathbf{A}) = -j\omega\Phi \quad \text{the generalized-Lorenz gauge} \quad (5)$$

with  $\mathbf{v} = 1/\mu$  as in our existing code.

**(2) and (3) are completely decoupled** — (2) contains no `A`, (3) contains no `Φ`. That is the whole point of this gauge: two independent systems instead of one coupled block system, with `E` recovered afterwards from (1). (2) is a scalar Helmholtz–Poisson equation; (3) is a vector Helmholtz equation carrying a gauge term.

**On  $\alpha$ .** Chew's §II.A allows  $\alpha$  to be a function of position and notes that all eigenvalues are positive for  $\alpha > 0$ , so (3) yields a positive-definite system. **\*\*He recommends  $\chi = \mu\epsilon^2$ , i.e.  $\alpha = 1$ \*\*, for the smallest condition number.**

**An inconsistency to resolve before coding.** §II.A defines  $\chi = \alpha\mu\epsilon^2$ , which is position-dependent through  $\mu$  and  $\epsilon$ . §II.C then says *"Assuming  $\chi = \mu_0\epsilon_0^2$ " and carries that constant into (21), (24) and (27). So the* **\*discretization as published uses a constant, free-space  $\chi$ \*\*, not the position-dependent one. Both are defensible — the gauge is a free choice — but they are different gauges and we must pick one deliberately. §8 lists this as an open decision.**

## Excitation

$$\nabla \cdot \mathbf{J} = -j\omega\rho \quad (13)$$

$$\nabla \cdot (\epsilon \nabla \Phi) + \chi \omega^2 \Phi = \nabla \cdot \mathbf{J} / (j\omega) \quad (14)$$

(14) is (2) with  $\rho$  eliminated through continuity, and is how the  $\Phi$ -equation is driven in a current-driven problem. Chew's remark is worth keeping: **(3) resembles the current–voltage relation of an inductor and (14) that of a capacitor** — the magneto-quasi-static and electro-quasi-static halves, solved separately and to the same accuracy.

## Boundary conditions

$$\Phi = \Phi_s \quad \mathbf{r} \in \Gamma_D \quad \text{Dirichlet} \quad (9)$$

$$\hat{\mathbf{n}} \cdot \epsilon \nabla \Phi = -\rho_s \quad \mathbf{r} \in \Gamma_N \quad \text{Neumann} \quad (10)$$

$$\hat{\mathbf{n}} \times \mathbf{A} = \mathbf{A}_s \quad \mathbf{r} \in \Gamma_D \quad \text{Dirichlet} \quad (11)$$

$$\hat{\mathbf{n}} \times (\mathbf{v} \nabla \times \mathbf{A}) = \mathbf{J}_s \quad \mathbf{r} \in \Gamma_N \quad \text{Neumann} \quad (12)$$

Chew notes that **with Whitney-0 for  $\Phi$  and Whitney-1 for  $\mathbf{A}$ , the conditions on the material interfaces are satisfied automatically** — which is the property that makes a mixed conductor/dielectric port tractable, and the reason this gauge was chosen. See [GAUGE\\_CHOICE.md](#) §2.2.

## Conductors

**\*\*Chew 2016 contains no  $\sigma$ .** Its conductors are PEC, imposed as boundary conditions; the volumes are dielectric. Finite conductivity enters the usual way, by folding it into a complex permittivity — in our convention

$$\epsilon_{\text{eff}} = \epsilon - j \sigma / \omega$$

so that  $j\omega \epsilon_{\text{eff}} \mathbf{E} = (\sigma + j\omega\epsilon) \mathbf{E} = \mathbf{J}_{\text{total}}$ . Everything below then holds with  $\epsilon \rightarrow \epsilon_{\text{eff}}$ . **This is not in the paper** and has one consequence worth flagging early:  $\chi = \alpha\mu\epsilon^2$  becomes complex, and whether the positive-definiteness argument of §II.A survives that is not established. §8.

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## 2. Why the gauge term cannot be discretized directly

The third term of (3) applies a divergence to  $\mathbf{A}$ . Discretized naively on Whitney-1 elements it **vanishes identically**, because

$$\nabla \cdot \omega_{mn} = \nabla \cdot ( \lambda_m \nabla \lambda_n - \lambda_n \nabla \lambda_m ) = 0$$

(15)

so a Ritz or Galerkin discretization of that term gives zero matrix entries and (3) silently degenerates to the ungauged double-curl equation.

But Whitney-1 elements are **not** solenoidal, because a constant gradient lies in their span:

$$\nabla \lambda_m = \sum_n \omega_{nm}$$

(16)

(the same fact that makes the tree-cotree null space exist at all, and the one Chew 2015 flags as the reason non-uniqueness survives curl-conforming elements).

**The resolution is an intermediate quantity.** Define

$$\mathbf{d} := (1/\chi) \nabla \cdot (\epsilon \mathbf{A})$$

and expand  $\mathbf{d}$  in Whitney-0 elements. Chew's Fig. 2 is the justification, and it is a differential-forms argument:

|              |                            |                       |                                |                                    |                                   |                                                       |
|--------------|----------------------------|-----------------------|--------------------------------|------------------------------------|-----------------------------------|-------------------------------------------------------|
| $\mathbf{A}$ | $\xrightarrow{* \epsilon}$ | $\epsilon \mathbf{A}$ | $\xrightarrow{- \nabla \cdot}$ | $\nabla \cdot \epsilon \mathbf{A}$ | $\xrightarrow{* \mu, * \epsilon}$ | $(1/\mu \epsilon^2) \nabla \cdot \epsilon \mathbf{A}$ |
| 1-form       |                            | 2-form                |                                | 3-form                             |                                   | 0-form                                                |

so  $\mathbf{d}$  lands in the same space as  $\Phi$ , exactly as (5) requires. Table I of the paper:

| QUANTITY                                              | SPACE                   | GEOMETRIC CONSTITUENT |
|-------------------------------------------------------|-------------------------|-----------------------|
| $\Phi$                                                | 0-form, potential space | primal nodes          |
| $\mathbf{A}$                                          | 1-form, field space     | primal edges          |
| $\mathbf{E}$                                          | 1-form, field space     | primal edges          |
| $(1/\mu \epsilon^2) \nabla \cdot \epsilon \mathbf{A}$ | 0-form, potential space | primal nodes          |

**No dual mesh is involved**, which is what keeps this compatible with an ordinary primal-mesh FEM code.

### 3. Expansions

$$\Phi = \sum_{n=1}^{N_n} \phi_n \lambda_n(\mathbf{r}) \quad \text{Whitney-0 (P1 nodal)} \quad (17)$$

$$\mathbf{A} = \sum_{n=1}^{N_e} \mathbf{a}_n \omega_n(\mathbf{r}) \quad \text{Whitney-1 (edge)} \quad (18)$$

$$\mathbf{d} = \sum_{n=1}^{N_n} \mathbf{d}_n \lambda_n(\mathbf{r}) \quad \text{Whitney-0 (P1 nodal)} \quad (19)$$

**$\Phi$  is P1 here. Ours is P2** — `src/element_matrix.cpp` builds a  $10 \times 10$   $\Phi$ - $\Phi$  block over 4 vertices plus 6 edge midpoints. The de Rham/Whitney argument above is what licenses the whole construction, and it is a statement about Whitney-0. Moving  $\Phi$  to P1 is the main discretization change this gauge asks for. §8.

Chew notes (20) that one could instead expand  $(1/\epsilon)\nabla \cdot \epsilon \mathbf{A}$ , but then the divergence theorem is needed and a surface integral term appears; he keeps (19) because it makes the third term of (3) integrable over the domain and yields a simple matrix representation.

### 4. Eliminating $\mathbf{d}$

Testing (19) with  $\lambda_m$  and using  $\chi = \mu_0 \epsilon_0^2$ :

$$\sum_n \langle \lambda_m, (1/\chi) \nabla \cdot (\epsilon \omega_n) \rangle \mathbf{a}_n = \sum_n \langle \lambda_m, \lambda_n \rangle \mathbf{d}_n \quad (21)$$

$$\mathbf{d} = \tilde{\mathbf{G}}^{-1} \mathbf{K}_{\text{NE}} \mathbf{a} \quad (22)$$

### 5. The matrices

All real; none of them carries the time convention.

$$[\tilde{G}]_{mn} = \int_{\Omega} \lambda_m \lambda_n d\Omega \quad (23)$$

$$[K_{NE}]_{mn} = \int_{\Gamma} (\epsilon/\chi) \lambda_m \omega_n \cdot \hat{n} d\Gamma - \int_{\Omega} (\epsilon/\chi) \nabla \lambda_m \cdot \omega_n d\Omega \quad (24)$$

$$[S1]_{mn} = \int_{\Omega} \nabla \lambda_m \cdot \epsilon \nabla \lambda_n d\Omega - \int_{\Gamma} \lambda_m \hat{n} \cdot \epsilon \nabla \lambda_n d\Gamma \quad (26)$$

$$[S2]_{mn} = \int_{\Omega} \chi \lambda_m \lambda_n d\Omega \quad (27)$$

$$[b_s]_m = \int_{\Omega} \lambda_m \rho d\Omega \quad (28)$$

$$[K1]_{mn} = \int_{\Omega} \nu (\nabla \times \omega_m) \cdot (\nabla \times \omega_n) d\Omega \quad (30)$$

$$[K2]_{mn} = \int_{\Omega} \epsilon \omega_m \cdot \omega_n d\Omega \quad (31)$$

$$[K_{EN}]_{mn} = \int_{\Omega} \epsilon \omega_m \cdot \nabla \lambda_n d\Omega \quad (32)$$

$$[b_k]_m = \int_{\Omega} \omega_m \cdot J d\Omega \quad (33)$$

## The two systems

$$\Phi\text{-equation:} \quad (S1 - \omega^2 S2) \phi = b_s \quad (25)$$

$$A\text{-equation:} \quad (K1 - \omega^2 K2 - K_{EN} \tilde{G}^{-1} K_{NE}) a = b_k \quad (29)$$

## Dimensions

|                     |                  |
|---------------------|------------------|
| $S1, S2, \tilde{G}$ | $N_n \times N_n$ |
| $K1, K2$            | $N_e \times N_e$ |
| $K_{EN}$            | $N_e \times N_n$ |
| $K_{NE}$            | $N_n \times N_e$ |
| $\phi, b_s$         | $N_n \times 1$   |
| $a, b_k$            | $N_e \times 1$   |

For a large 3-D mesh  $N_n/N_e \approx 1/7.3$ , so the A-system costs about **13.7 %** more unknowns than an E formulation. Chew measures 15.04 % and 15.35 % on his two examples.

## Static bounds, for preconditioning and for sanity

$$\begin{aligned} K &= K1 - \omega^2 K2 - K_{EN} \bar{G}^{-1} K_{NE} & K' &= K1 - K_{EN} \bar{G}^{-1} K_{NE} & (35), (37) \\ S &= S1 - \omega^2 S2 & S' &= S1 & (34), (36) \end{aligned}$$

$K'$  and  $S'$  are the  $\omega = 0$  limits and are **frequency-independent**. Chew's reported condition numbers:  $K$  stays bounded by  $K'$  at  $5.94e5$  and  $1.21e6$  across his two cases and the whole frequency range, against the E formulation's  $1.04e17$  at 10 kHz.

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## 6. Solving — direct is explicitly supported

Worth recording plainly, because the first reading of this paper suggested otherwise. §II.D: *sparse direct solvers such as UMFPACK and CHOLMOD are suitable candidates* for problems of tens of thousands of unknowns, and §III states that **UMFPACK multifrontal LU is applied first** to make the solution as accurate as possible. The ILU/GMRES material is an optimization for wideband sweeps — the frequency-independent  $K'$  can be factored once and reused at every frequency — **not a requirement of the method**.

So this composes with our MUMPS direct path rather than fighting it.

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## 7. Avoiding the sparse approximate inverse — our variant

**This section is my derivation. It is not in either paper and has not been tested.** It must be checked symbolically and numerically before it is relied on. Everything above is transcribed; everything here is not.

$\bar{G}^{-1}$  is the problem.  $\bar{G}$  is the nodal mass matrix, so  $\bar{G}^{-1}$  is dense, and  $K_{EN} \bar{G}^{-1} K_{NE}$  formed explicitly would destroy sparsity. Chew's answer is a **sparse approximate inverse** — on his examples  $\bar{G}^{-1}$  is replaced by an SAI with 9057 and 7147 nonzeros, 98.63 % and 99.98 % sparse. That is an approximation adopted to keep the system  $N_e \times N_e$  for iterative solving.

We factor directly, so we can avoid it entirely by keeping  $\mathbf{d}$  as unknowns. From (22),  $\tilde{\mathbf{G}} \mathbf{d} = \mathbf{K}_{NE} \mathbf{a}$ , and (29) becomes  $(\mathbf{K}_1 - \omega^2 \mathbf{K}_2) \mathbf{a} - \mathbf{K}_{EN} \mathbf{d} = \mathbf{b}_k$ :

$$\begin{bmatrix} \mathbf{K}_1 - \omega^2 \mathbf{K}_2 & -\mathbf{K}_{EN} \\ \mathbf{K}_{NE} & -\tilde{\mathbf{G}} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{d} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_k \\ \mathbf{0} \end{bmatrix}$$

Everything in it is sparse and exact. The cost is  $+Nn$  unknowns — about 14 % on top of the A-system, i.e. roughly the same again as the formulation already costs over an E formulation.

**It can be made symmetric.** Ignoring the surface term, (24) is

$$[\mathbf{K}_{NE}]_{mn} = -(1/\chi) \int_{\Omega} \epsilon \nabla \lambda_m \cdot \omega_n \, d\Omega = -(1/\chi) [\mathbf{K}_{EN}]_{nm}$$

so  $\mathbf{K}_{NE} = -(1/\chi) \mathbf{K}_{EN}^T$ . Scaling the second block row by  $\chi$ :

$$\begin{bmatrix} \mathbf{K}_1 - \omega^2 \mathbf{K}_2 & -\mathbf{K}_{EN} \\ -\mathbf{K}_{EN}^T & -\chi \tilde{\mathbf{G}} \end{bmatrix} \begin{bmatrix} \mathbf{a} \\ \mathbf{d} \end{bmatrix} = \begin{bmatrix} \mathbf{b}_k \\ \mathbf{0} \end{bmatrix} \quad \text{symmetric}$$

which is what MUMPS's complex-symmetric path wants.

**Three things to check before trusting this:**

1. **The surface term of (24)** —  $\int_{\Gamma} (\epsilon/\chi) \lambda_m \omega_n \cdot \hat{n} \, d\Gamma$  — is dropped above. It must either vanish under our boundary conditions or be carried, and if carried it breaks the exact symmetry.
2. **\*\* $\chi$  inside the scaling\*\*** must be constant for the row scaling to be a congruence. If we adopt a position-dependent  $\chi$  (§1), this derivation needs redoing.
3. **Conditioning.** The block system is larger and has a zero-ish structure in the  $(2,2)$  corner; whether its condition number behaves as well as Chew's reported  $\mathbf{K}$  is an empirical question, and is the first thing the spike should measure.

If any of these fails, Chew's SAI route remains available and is the fallback.

## 8. Open decisions

|                                                 |                                                                                                                                                                                                                                                                               |
|-------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ** $\chi$ constant or position-dependent?<br>** | §II.A defines $\chi = \alpha\mu\epsilon^2$ ; §II.C discretizes with $\chi = \mu_0\epsilon_0^2$ constant. Constant makes §7's symmetrization a clean congruence. Position-dependent is the "generalized" gauge as stated. <b>Pick deliberately and record why.</b>             |
| $\Phi: P2 \rightarrow P1?$                      | The Whitney/de Rham argument is about Whitney-0. Our $\Phi$ is P2. Either move $\Phi$ to P1 or redo the compatibility analysis for P2 — the latter is research, not implementation.                                                                                           |
| Lossy conductors                                | Fold $\sigma$ into $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$ . Not in the paper. Makes $\chi$ complex; the positive-definiteness argument of §II.A is then unproven.                                                                                                |
| ** $\bar{G}^{-1}$ : block system or SAI? **     | §7 versus Chew's SAI. Decide on measured conditioning, not on preference.                                                                                                                                                                                                     |
| Per-region boundary conditions on a port face   | Needed for the mixed port, which is the acceptance test (§10). The formulation supplies it through the $\Gamma_D / \Gamma_N$ partition of (9)–(12); our parser rejects per-surface BCs today. Parser and binding work, independent of the gauge, and can proceed in parallel. |

## 9. What maps onto existing code

From `include/aphi_solver/element_matrix.hpp`, we already build per-tet:

| OURS                                                                                                         | THEIRS                    | STATUS                                                                              |
|--------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------------------------------|
| A-A curl part, $\mathbf{v} \int (\nabla \times \mathbf{W}_i) \cdot (\nabla \times \mathbf{W}_j)$             | ** K1 ** (30)             | identical                                                                           |
| A-A mass part, $\alpha \int \mathbf{W}_i \cdot \mathbf{W}_j$ , $\alpha = j\omega\sigma - \omega^2\epsilon$   | ** $-\omega^2 K2$ ** (31) | same shape; our $\alpha$ already carries $\sigma$ via the complex- $\epsilon$ route |
| A- $\Phi$ coupling, $\beta \int \mathbf{W}_i \cdot \nabla \mathbf{S}_b$ , $\beta = \sigma + j\omega\epsilon$ | ** K_EN ** (32)           | same shape, different coefficient and $\Phi$ space                                  |
| $\Phi$ - $\Phi$ , $\beta \int \nabla \mathbf{S}_a \cdot \nabla \mathbf{S}_b$                                 | ** S1 ** (26)             | same shape, coefficient $\epsilon$ not $\beta$                                      |

### New work:

- \*\*  $\bar{G}$  \*\* (23) — nodal mass. Trivial; the quadrature exists.

- **S2** (27) — nodal mass weighted by  $\chi$ . Trivial given  $\bar{G}$ .
- **K\_NE** (24) — the genuinely new one. A volume term that is  $K_{EN}^T$  up to the  $\epsilon/\chi$  weighting, plus a **surface** term over  $\Gamma$ . Our assembly has no surface-integral path for this block today.
- **The gauge term's coupling** — either the block system of §7 or an SAI.
- **Decoupling the solve** — (25) and (29) are independent, so the driver changes from one factorization to two.
- **Removing tree–cotree** from the A path.

**What goes away:** `tree_cotree.cpp`, `gauge_variants.cpp` and the gauging in `dof_map.cpp` cease to be load-bearing. Keep `07_GaugeInvariance` — it becomes the test that  $\Phi$  *no longer* moves.

## 10. Validation plan

### The acceptance test is the mixed-material port

**This is why the gauge is being replaced at all.** `ROADMAP.md` Phase 03 names a port face with conductor on one side and dielectric on the other, inside the domain, as the reason tree–cotree is inadequate; it is corroborated by the project owner's experience of a commercial solver running such a port to completion and returning wrong answers (`GAUGE_CHOICE.md` §1.3). Everything else in this list is a necessary condition. **This one is the goal**, and the gauge has not succeeded until it passes.

**The formulation already provides what a mixed port needs**, which the first draft of this section under-stated. Eqs (9)–(12) partition the boundary into  $\Gamma_D$  and  $\Gamma_N$  **separately for  $\Phi$  and for  $A$** . A mixed port face is exactly that partition:

| PART OF THE FACE | $\Phi$                                                        | $A$                                                         |
|------------------|---------------------------------------------------------------|-------------------------------------------------------------|
| conductor        | $\Phi = \Phi_S$ — Dirichlet (9)                               | $\hat{n} \times A = A_S$ — Dirichlet (11)                   |
| dielectric       | $\hat{n} \cdot \epsilon \nabla \Phi = -\rho_S$ — Neumann (10) | $\hat{n} \times (\nu \nabla \times A) = J_S$ — Neumann (12) |

and Chew's remark that with Whitney-0  $\Phi$  and Whitney-1  $A$  the **interface conditions are satisfied automatically** is what makes the conductor/dielectric boundary inside the face unproblematic.

**The gap is on our side and it is bounded.** `src/input_file.cpp` currently fails with "*per-surface boundary conditions are not supported yet*", and every port type we have imposes a single potential over the whole tagged surface. So a mixed port needs **per-region boundary conditions on a port face** — Dirichlet on the conductor part, natural on the dielectric part. That is parser and binding work, not a reformulation, and it is independent enough to proceed in parallel with the gauge.

### The baseline cannot be built yet, and that sets the order

**A mixed port is not expressible by this solver today.** `[boundary]` "takes no name and applies to the whole outer boundary" — per-surface conditions are explicitly unimplemented — and every port type imposes one potential over the whole tagged surface. So building the coax now would produce one of two useless results:

- tag the whole cross-section as one port → the dielectric is shorted to the conductor's potential → the ill-posed garbage the first case 08 attempt produced;
- tag only the conductors → it is not a mixed port, and case 05 has already measured what an interior conductor cut does.

**Either way it would measure the port model's inability to pose the problem, not the gauge's failure to solve it** — the same confound that wrecked the first attempt, on a larger mesh. "Measure the failure before fixing it" does not rescue it: a failure cannot be measured on a problem that cannot be stated.

So the dependency is strict rather than parallel:

per-region port BCs    ->    coax baseline    ->    mixed-port acceptance test

**The first case 08 attempt is worth remembering here.** A straight rod with an internal cut has no return path, so the system was ill-posed; both the mixed face and the conductor-only control came back wrecked, and it was the *control* that showed the geometry was at

fault rather than the gauge. Any rebuild needs a closed circuit — coax, up the centre and back through the shield.

## In order

Steps 1 and 2 are genuinely independent and can run in parallel. Nothing after them can start early.

1. **Spike the gauge term** — assemble `K_NE` and  $\tilde{G}$  on `07_GaugeInvariance`'s 8,365-tet mesh. Is the §7 block system complex-symmetric, does MUMPS factor it, and how does its condition number compare with the tree-gauged system? **No new geometry, no port work, commits to nothing, and it retires the largest unknown in the plan.**
2. **Per-region boundary conditions on a port face** — Dirichlet on the conductor part, natural on the dielectric part, shaped by the `r_D / r_N` partition of (9)–(12). Parser and binding work, not reformulation. **On the critical path regardless of the gauge**, because without it there is no acceptance test.
3. **Case 08 built and failing** — the coax mixed port, demonstrated broken under tree–cotree, with numbers. Nearly free once step 2 exists.
4. **Reproduce Chew's Fig. 5** qualitatively — condition number flat against frequency where the E formulation blows up. His `5.94e5` and `1.21e6` are the target order.
5. **\*\* 07\_GaugeInvariance inverted\*\*** —  $\Phi$  must now be mesh-independent. The permutation harness stays; the *potential-moves* assertions flip.
6. **Case 05's local field** — the 8 % tree-dependence of `J` within 2 mm of the cut (`GAUGE_CHOICE.md` §1.3) must go away. A necessary condition, and the only one measurable on an existing case today.
7. **Cases 01–06 unchanged** on every field quantity. This is a reformulation, not a change of physics.
8. **The mixed port passes** — step 3's baseline, now correct. **Tree-independence is not the test here**: once the tree is gone there is no tree to depend on, so the standard is that the local field on and near the port face **converges under mesh refinement**.

# 11. Spike results — measured, and they change §7 and §8

`tools/gauge_spike/` assembles §5's matrices on 518 tets and computes exact condition numbers by dense SVD. It ran before any C++ was written, which is the only reason the findings below cost a day rather than a month.

## The formulation does what Chew claims — for dielectrics

His own materials: dielectric everywhere, no  $\sigma$ .

| FREQ (HZ) | COND GAUGED | COND UNGAUGED |
|-----------|-------------|---------------|
| 1e0       | 2.492e+02   | 1.884e+18     |
| 1e4       | 2.492e+02   | 2.584e+14     |
| 1e10      | 3.907e+02   | 5.729e+02     |

**Flat across ten decades** while the ungauged system degrades by sixteen orders as  $\omega \rightarrow 0$ . That is Fig. 5 of the paper reproduced in character, and it is what validates the assembly — every other number here depends on it.

## It does not survive $\sigma$ folded into $\epsilon$

| FREQ (HZ) | COND GAUGED | COND UNGAUGED | E CONTRAST |
|-----------|-------------|---------------|------------|
| 1e0       | 2.046e+22   | 1.655e+06     | 2.317e+17  |
| 1e4       | 9.272e+17   | 5.745e+02     | 2.317e+13  |
| 1e10      | 1.527e+10   | 1.118e+09     | 2.317e+07  |

The condition number **tracks the  $\epsilon$  contrast** at about `contrast × 1e4`, and below roughly 1 GHz **the gauged system is worse than the ungauged one.**

Not a bug:  $\epsilon_{\text{eff}} = \epsilon - j\sigma/\omega$  gives  $\sigma/\omega\epsilon = 2.3e17$  between copper and dielectric at 1 Hz. **Chew has no finite conductivity** — PEC boundary conditions, dielectric volumes — so his  $\epsilon$  varies by 4.5.  $\epsilon_{\text{eff}}$  is §1's addition, and it is what breaks the conditioning.

**This qualifies the roadmap's low-frequency rationale, sharply.** Phase 03 wants this gauge partly to remove the low-frequency floor. For a *dielectric* problem it does, completely. For **lossy metal at low frequency it makes conditioning worse**, monotonically in  $\sigma/\omega\epsilon$ .

It does not invalidate the choice — we want this gauge for the **embedded interface conditions** at a mixed port (§10), which the above does not touch. But "fixes low-frequency breakdown" must not be claimed for conductors without qualification.

**At THz the problem recedes.** The contrast falls as  $1/\omega$ :  $2.3e7$  at 10 GHz and about  $2.3e5$  at 1 THz, where conditioning should be unremarkable. The difficulty is specific to the low-frequency end with good conductors.

## $\chi$ must be position-dependent — and that breaks §7's symmetry

COND AT 1 HZ

$\chi = \mu_0 \epsilon_0^2$  constant — his §II.C

$7.026e+37$

$\chi = \mu \epsilon^2$  per element — his §II.A

$2.046e+22$

Fifteen orders, because  $\epsilon^2/\chi = 1/\mu = v$  self-normalizes to the curl–curl coefficient  
 \*\*only when  $\chi$  carries the local  $\epsilon$ . §8's open question is therefore answered: position-dependent\*\*.

But §7's symmetrization is a global row scaling by  $\chi$ , a congruence only for constant  $\chi$ . Measured per element,  $K_{NE}$  is no longer a scalar multiple of  $K_{EN}^T$  — residual **O(1)**, not  $1e-16$ .

**As derived, the block system is symmetric or well-conditioned, not both.** The options are a per-element scaling that is still a congruence, an unsymmetric system, or Chew's SAI route.

\*\*Qualified by `GAUGE_CHOICE.md` §11.4, added after this section.\*\* Balian et al. (2023) give the congruence explicitly — scale the equation *and* the unknown by the same  $(\sigma + j\omega\epsilon)^{(-1/2)}$ , so symmetry survives. It does **not** fully rescue us:  $\epsilon/\chi$  sits inside the element integral, so a diagonal congruence reproduces  $K_{NE} = -c \cdot K_{EN}^T$  for every node interior to one material and **fails at material-interface nodes** — exactly where a mixed port lives. The literature's answer is to assign interface DOFs to the conductor block and scale blockwise, rather than to seek exactness.

## What §7 got right, and what it got wrong

**Right:** with constant  $\chi$  the block system is exactly complex-symmetric — measured  $1e-21$  to  $8e-17$ . The derivation  $K_{NE} = -(1/\chi)K_{EN}^T$  holds.

**Wrong:** §7 ignored scaling.  $\chi \approx 1e-28$  puts the (2,2) block  $1e40$  below the (1,1) block, and the first spike run reported  $1e46$  because of it. The scaling  $c$  is free — it cancels in the Schur complement — so it can be chosen for conditioning, and  $c^2 = \|K_{uu}\| / (\chi \|G\|)$  recovered 26 orders. **Any block form must be scaled deliberately**; the naive one is unusable.

**Why the controls are the whole point**

The first two spike runs concluded the formulation was unusable, at  $1e38$ . Both were wrong — the first was my block scaling, the second was finite conductivity, which is not in the paper. Only running **Chew's materials unchanged** showed the assembly had been right the whole time.

Without that control this document would now be recommending against a sound method.

**Revised open questions**

|                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| **How to handle $\sigma$ **           | $\epsilon_{eff}$ is measured to wreck conditioning below ~1 GHz. <b>This is a known, published phenomenon with published remedies</b> — see GAUGE_CHOICE.md §11, added after this section was written. Balian et al. (2023) name the same $\sigma/\omega\epsilon$ mechanism and fix it by scaling equations <i>and</i> unknowns analytically before assembly; Herles et al. (2025) use a <b>different gauge condition inside conductors than outside</b> , which is the "different $\chi$ inside conductors" line below, built and measured. Not a blocker. |
| **Symmetry with per-element $\chi$ ** | As above. Symmetric or well-conditioned, not both, as currently derived.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| <b>The surface term of (24)</b>       | Still omitted in the spike. Must vanish under our boundary conditions or be carried; carrying it may break symmetry independently of $\chi$ .                                                                                                                                                                                                                                                                                                                                                                                                               |